

MAESTRO INSTANCE SETUP IN MICROSOFT AZURE

This document provides the information on the configuration of an instance in Azure Cloud for on premise installation of Maestro.

The described setup is required to enable necessary access and performance for the Maestro application.

1.1 INSTANCE CONFIGURATION

- **Minimum instance type:** Standard B2ms (2 vCPU, 8 GB RAM)
- **Disk size:** 100 GB
- **Operating System:** Ubuntu 22.04

1.2 INSTANCE SETUP FLOW

To perform the setup steps below, use Azure CLI.

1. Login to Azure CLI and specify subscription to use for Maestro deployment:

```
export SUBSCRIPTION_ID=""  
az account set -s $SUBSCRIPTION_ID
```

Specify the following variables:

```
export REGION=""  
export PATH_TO_SSH_KEY=""  
export PREFIX=""  
export ACCESS_IP_ADDRESS_LIST=""  
export ADMIN_IP_ADDRESS_LIST=""
```

where:

- **REGION** – the region to deploy resources to (e.g. REGION="eastus2")
- **PATH_TO_SSH_KEY** – the path to the public ssh for ssh login to instance (e.g. PATH_TO_SSH_KEY="/home/ubuntu/my_ssh_key.pub ")
- **PREFIX** – will be used for resource naming. Prefix also will be used for Maestro DNS name generation (e.g. PREFIX="maestro-demo")
- **ACCESS_IP_ADDRESS_LIST** – the list of IP addresses separated by spaces that will be used to allow user access to Maestro (e.g. ACCESS_IP_ADDRESS_LIST ="1.1.1.0/24 2.2.2.0/24")
- **ADMIN_IP_ADDRESS_LIST** – the list of IP addresses separated by spaces that will be used to allow access for administrators to configure Maestro (e.g. ADMIN_IP_ADDRESS_LIST ="3.3.3.0/24 4.4.4.0/24")

2. Create a resource group:

```
az group create --location $REGION --name "$PREFIX-rg"
```

3. Create a virtual instance with the necessary capacities and configurations:

```
az vm create -n "$PREFIX-vm" -g "$PREFIX-rg" --image Ubuntu2204 --
admin-username ubuntu --os-disk-size-gb 100 --storage-sku
StandardSSD_LRS --size Standard_B2s --ssh-key-values
$PATH_TO_SSH_KEY --public-ip-address "$PREFIX-public-ip" --public-
ip-address-allocation static
```

4. Add a DNS name:

```
az network public-ip update -g "$PREFIX-rg" -n "$PREFIX-public-ip" -
-dns-name "$PREFIX" --allocation-method Static
```

5. Open the ports necessary for Maestro on premise setup (22, 80, 443, 3000, 8080, 8200, 9000, 9001, 9090, 15672, 27017):

```
az network nsg rule create --resource-group "$PREFIX-rg" --nsg-name
"$PREFIX-vmNSG" --name Allow-22 --priority 2000 --source-address-
prefixes $ADMIN_IP_ADDRESS_LIST --destination-port-ranges 22 --
direction Inbound --access Allow --protocol Tcp

az network nsg rule create --resource-group "$PREFIX-rg" --nsg-name
"$PREFIX-vmNSG" --name Allow-443 --priority 2002 --source-address-
prefixes $ACCESS_IP_ADDRESS_LIST --destination-port-ranges 443 --
direction Inbound --access Allow --protocol Tcp

az network nsg rule create --resource-group "$PREFIX-rg" --nsg-name
"$PREFIX-vmNSG" --name Allow-3000 --priority 2003 --source-address-
prefixes $ACCESS_IP_ADDRESS_LIST --destination-port-ranges 3000 --
direction Inbound --access Allow --protocol Tcp

az network nsg rule create --resource-group "$PREFIX-rg" --nsg-name
"$PREFIX-vmNSG" --name Allow-8080 --priority 2004 --source-address-
prefixes $ADMIN_IP_ADDRESS_LIST --destination-port-ranges 8080 --
direction Inbound --access Allow --protocol Tcp

az network nsg rule create --resource-group "$PREFIX-rg" --nsg-name
"$PREFIX-vmNSG" --name Allow-8200 --priority 2005 --source-address-
prefixes $ADMIN_IP_ADDRESS_LIST --destination-port-ranges 8200 --
direction Inbound --access Allow --protocol Tcp

az network nsg rule create --resource-group "$PREFIX-rg" --nsg-name
"$PREFIX-vmNSG" --name Allow-9000 --priority 2006 --source-address-
prefixes $ADMIN_IP_ADDRESS_LIST --destination-port-ranges 9000 --
direction Inbound --access Allow --protocol Tcp

az network nsg rule create --resource-group "$PREFIX-rg" --nsg-name
"$PREFIX-vmNSG" --name Allow-9001 --priority 2007 --source-address-
```

```
prefixes $ADMIN_IP_ADDRESS_LIST --destination-port-ranges 9001 --  
direction Inbound --access Allow --protocol Tcp  
  
az network nsg rule create --resource-group "$PREFIX-rg" --nsg-name  
"$PREFIX-vmNSG" --name Allow-9090 --priority 2008 --source-address-  
prefixes $ADMIN_IP_ADDRESS_LIST --destination-port-ranges 9090 --  
direction Inbound --access Allow --protocol Tcp  
  
az network nsg rule create --resource-group "$PREFIX-rg" --nsg-name  
"$PREFIX-vmNSG" --name Allow-15672 --priority 2009 --source-address-  
prefixes $ADMIN_IP_ADDRESS_LIST --destination-port-ranges 15672 --  
direction Inbound --access Allow --protocol Tcp  
  
az network nsg rule create --resource-group "$PREFIX-rg" --nsg-name  
"$PREFIX-vmNSG" --name Allow-27017 --priority 2010 --source-address-  
prefixes $ADMIN_IP_ADDRESS_LIST --destination-port-ranges 27017 --  
direction Inbound --access Allow --protocol Tcp
```